

A Report for TD619: Energy Policy and Planning On Solar Mini-grid Design Using a Multidimensional Approach

Course Project Report for TD 619

Submitted by

Arrunraaj K (24M1345)

Yash Bhongade (24M1346)

Yaswanth B (24M1349)

Ravneet Singh (24M1350)



Under the guidance of Prof. Anand B. Rao Centre for
Technology Alternatives for Rural Areas (C-TARA) Indian
Institute of Technology Bombay

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DECLARATION

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Arrunaaj K (24M1345)

Yash Bhongade (24M1346)

Yaswanth B (24M1349)

Ravneet Singh (24M1350)

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Chapter 1

Introduction

In this 21st Century, access to electricity is a basic need for economic growth, health, education and overall quality of life. Yet, many remote villages in India does not have grid connection due to geography, high initial investment for low population density. For such cases solar photovoltaic (PV) plays a vital role in generating affordable electricity compared with wind energy. This report explores the design of a solar PV mini-grid system for a village without grid access, focusing on aspects such as electric load analysis, solar irradiance analysis, system sizing, component selection, and economic feasibility. The proposed mini grid will integrate solar PV and battery storage for households in the Rajapuri village, Thane district, Maharashtra. Installing solar PV system in Rajapuri village will help to improve the living standards of people, supporting local development.

The following report outlines the Techno-Economic Design of mini grid, and social dimension, institutional dimension, environmental dimension of installing mini grid in the Rajapuri village, Thane district, Maharashtra

Chapter 2

Solar Mini-grid Techno-Economic Design

2.1 Solar Mini-Grid Technical Design

The electrical load data analysis for the given time period (July 2018 – May 2024) shows that there are fluctuations in the data after 500 hours (from 2020 onwards) and after installing the system, load is consistently increasing, therefore for designing of the mini grid system, 2019 electrical load data is taken, which is consistent and it is only one year after the installation of the system.

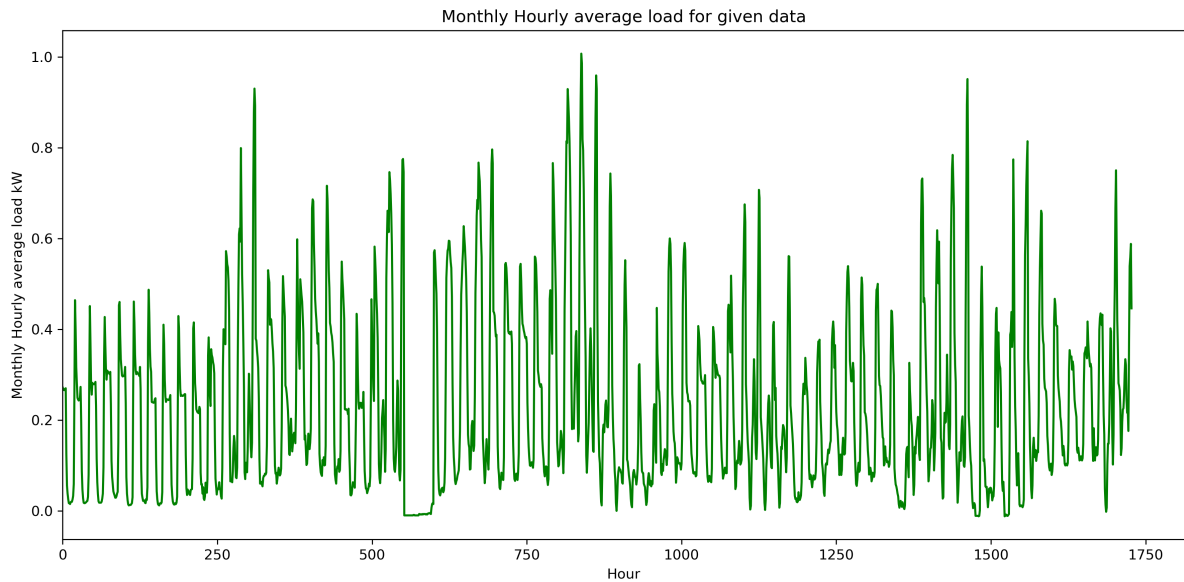


Figure 2.1: Monthly hourly average data for the given time period

Assumptions:

- Capacity factor = 18%
- No system breakdown
- Battery capacity only to supply night time consumption

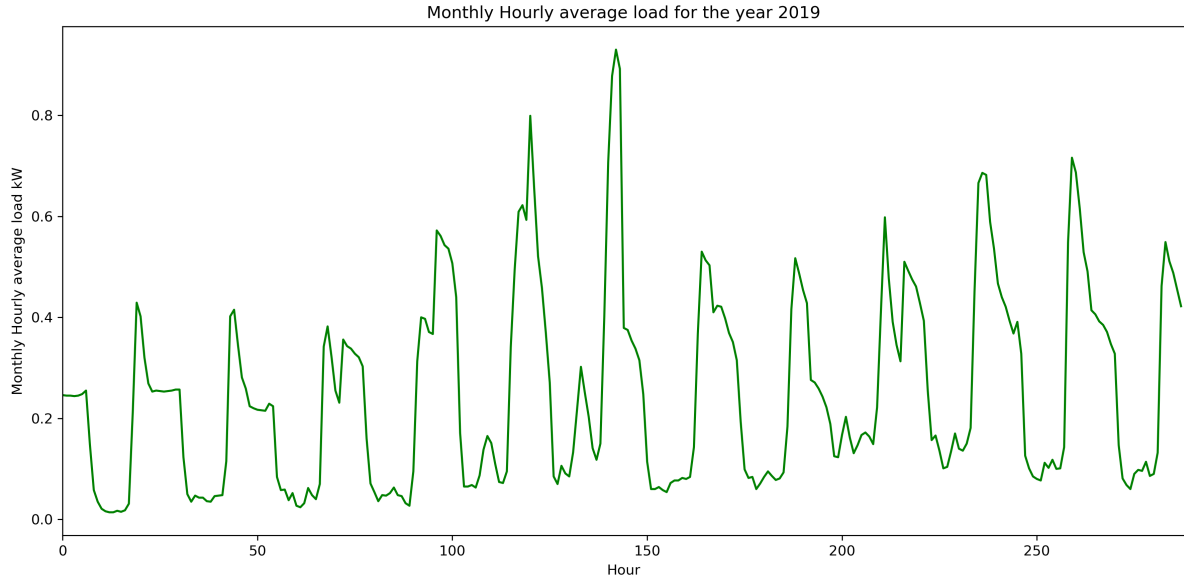


Figure 2.2: Monthly hourly average data for the year 2019

- No power degradation
- Inverter capacity based on the peak load only
- Battery efficiency = 90%
- Depth of discharge (DoD) for lead acid battery = 50%
- Designing the system for max load and max energy consumption

Table 2.1: Data analysis

S. No.	Parameter	Value
1	Max hourly load	0.930 kW
2	Min hourly load	0.014 kW
3	Average daily load	0.5-0.8 kW
4	Day time unit consumption	4.5- 6 kWh
5	Night time unit consumption	3 - 5 kWh
6	Daily unit consumption	7-12 kWh/ Day

Calculations:

To satisfy the 0.93 kW peak load, a 5.2 kW solar panel system is proposed based on 18% capacity factor.

$$\text{Daily Energy Generation} = 5.2 \times 0.18 \times 24 = 22.464 \text{ kWh}$$

$$\text{Remaining Energy After Day Load (6 kWh)} = 22.464 - 6 = 16.464 \text{ kWh}$$

Battery Sizing:

$$\begin{aligned}\text{Battery Capacity} &= \frac{\text{Night-time Consumption}}{\text{DoD} \times \text{Battery Efficiency}} \\ &= \frac{5}{0.5 \times 0.9} = 11.11 \text{ kWh}\end{aligned}$$

Since the remaining energy after day-time load is greater than the required battery capacity, the battery can be fully charged without needing additional solar panel capacity.

Inverter Sizing: To supply a maximum load of 0.93 kW, an inverter with at least 1 kW capacity is selected.

Design Specifications:

Table 2.2: Design summary

S. No.	Parameter	Value
1	Solar Panel Capacity	5.2 kW
2	Battery Capacity	11.5 kW
3	Inverter Capacity	1 kW

2.2 Mini grid economics

Solar Panel System Cost Analysis:

Parameters:

- Cost of Solar panel 1 kW capacity after subsidy = ₹30000
- Battery 1 kW h capacity cost = ₹15000
- Number of batteries required = 12 (11 - 1 kW h and 1 - 0.5 kW h)
- Operating and maintenance cost per battery per year = ₹200
- Inverter and Battery replacement after 15 and 10 years respectively
- Cost of 1 kW Inverter = ₹50000
- Lifetime of the system = 25 years
- Discount rate = 7%
- No salvage value considered

Present Value Calculations:

$$\text{Initial investment: } PV_0 = C_0 = Rs.378500$$

$$\begin{aligned}\text{Operating cost: } PV_A &= A \times \left(\frac{1 - (1 + r)^{-N}}{r} \right) \\ &= 2400 \times \left(\frac{1 - (1.07)^{-25}}{0.07} \right) = Rs.27968.64\end{aligned}$$

$$\text{Battery replacement at 10 years: } PV_{10} = \frac{C_{10}}{(1 + r)^{10}} = \frac{172500}{(1.07)^{10}} = \frac{172500}{1.9672} = Rs.87688.59$$

$$\text{Battery replacement at 20 years: } PV_{20} = \frac{C_{20}}{(1 + r)^{20}} = \frac{172500}{(1.07)^{20}} = \frac{172500}{3.8697} = Rs.44576.58$$

$$\text{Inverter replacement at 15 years: } PV_{15} = \frac{C_{15}}{(1 + r)^{15}} = \frac{50000}{(1.07)^{15}} = \frac{50000}{2.7590} = Rs.18122.51$$

$$\text{Net present value: } PV_{\text{total}} = PV_0 + PV_A + PV_{10} + PV_{15} + PV_{20} = Rs.556856.32$$

Equivalent Annual Cost (EAC):

$$\begin{aligned}\text{EAC} &= PV_{\text{total}} \times \left(\frac{r(1 + r)^N}{(1 + r)^N - 1} \right) \\ &= 556856.32 \times \left(\frac{0.07(1.07)^{25}}{(1.07)^{25} - 1} \right) \\ &= 556856.32 \times 0.0858 = Rs.47778.27\end{aligned}$$

Annual Electricity Generation:

$$\text{Annual Electricity generation} = 5.2 \times 0.18 \times 8760 = 8199.36 \text{ kWh}$$

Homer results: Homer Pro results are shown in the Figure 2.3 and Figure 2.4

Table 2.3: Homer results

S. No.	Parameter	Value
1	Solar panel capacity	10.4 kW
2	Battery capacity	22 kW
3	Inverter	1.12 kW
4	Discount rate	5.2% kWh
5	Levelized cost	38.53 per kWh

2.3 Comparison of the Design with the Existing System

Compared to the current system, the manual design seems less expensive on initial consideration. HOMER Pro's method, however, fundamentally differs by utilizing real weather data and optimizing the system to support peak demand times. The software

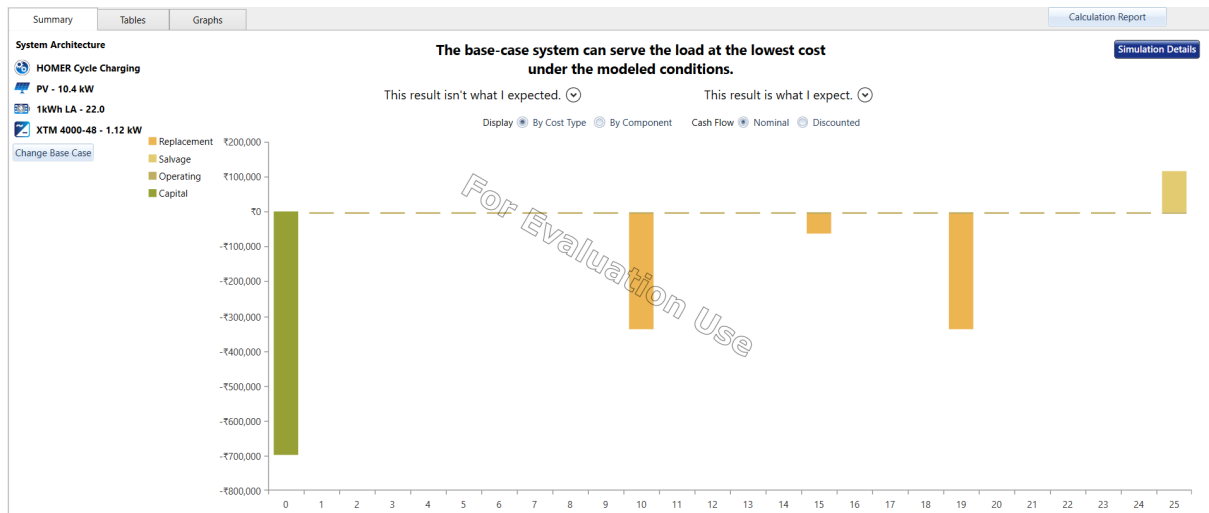


Figure 2.3: Homer cash flow diagram

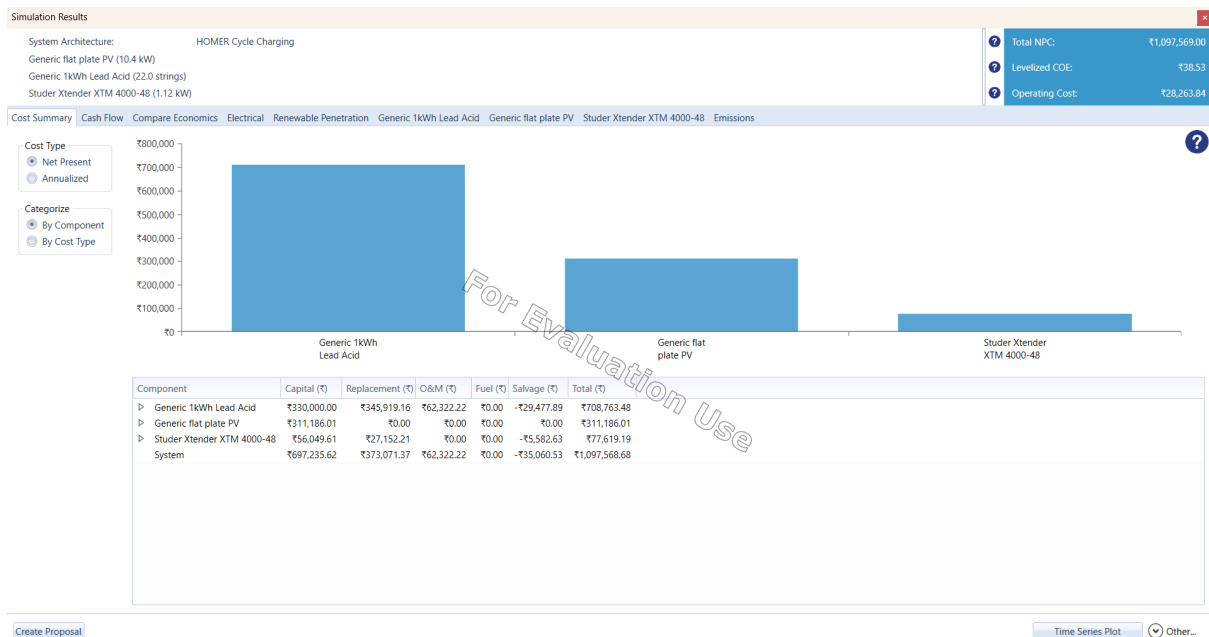


Figure 2.4: Homer cost summary

Table 2.4: Comparison of System Parameters

Parameters	Existing System	Manual Design	Homer Pro
Solar panel capacity	6.08 kW	5.2 kW	10.4 kW
Battery capacity	16 kW h	11.5 kW h	22 kW h
Inverter capacity	3 kW	1 kW	1.12 kW

gives top priority to full, unbroken power supply all year round, and that is the reason it suggests much higher capacities of batteries and solar. While the manual design does not consider the real time weather data, the configuration of HOMER Pro ensures higher reliability by always considering variability in real-world weather and a full load.

Chapter 3

Social Dimension

Access to electricity is transformative for rural communities like Rajapuri village. A solar mini-grid will empower residents through multidimensional benefits:

- **Improving Quality of Life:** Reliable electricity enables lighting, fans, and mobile charging, eliminating dependence on kerosene lamps and improving indoor air quality [3].
- **Boosting Education:** Children gain extended study hours after dark, while schools can integrate digital tools, significantly enhancing learning outcomes [6].
- **Supporting Livelihoods:** Small businesses (e.g., tailoring, milling) extend operating hours, while agro-processing and refrigeration dramatically reduce post-harvest losses [3].
- **Enhancing Healthcare:** Clinics can refrigerate vaccines and power medical devices, strengthening community health resilience [3].

Key Success Factors

- **Community Engagement:** Training locals in system maintenance fosters ownership and ensures sustainability.
- **Transparent Tariffs:** Fair pricing structures guarantee affordability for all households.
- **Gender Inclusion:** Womens self-help groups can manage revenue collection, creating economic opportunities.

Chapter 4

Institutional dimension

Critical Success Factors

- **Policy Support:** Government interventions like 30% capital cost subsidies and renewable purchase obligations significantly reduce financial barriers [8].
- **Local Governance:** Village Energy Committees (VECs) should oversee daily operations, backed by technical support from NGOs or state agencies [4].
- **Strategic Partnerships:**
 - Collaboration with DISCOMs for future grid interconnection
 - Private operators for specialized maintenance
- **Capacity Building:** Training programs covering system troubleshooting and financial literacy ensure community ownership [1].

Implementation Challenges & Solutions

- **Bureaucratic Hurdles:** Permit delays can be mitigated through:
 - Single-window clearance systems
 - Pre-approved technical designs
- **Skill Gaps:** Invest in vocational training centers for:
 - Solar technician certification
 - Basic financial management

Chapter 5

Environmental Dimension

- **Energy Source Selection**
 - Solar, wind, hydro, or hybrid systems have varying footprints
 - Renewable sources generally greener than diesel but require lifecycle analysis [2]
- **Key Impact Factors**
 - Resource extraction methods (e.g., rare earth metals for solar panels)
 - Waste management across system lifespan (especially batteries)
 - Land use changes during installation [2]
- **Environmental Impact Assessments (EIAs)**
 - Essential for detecting threats early (World Bank 2020 recommendation)
 - Identifies risks to local ecosystems and resources
 - Addresses pollution risks during construction/operation
- **Documented Challenges**
 - Vegetation clearance for solar projects (African development bank, 2019)
 - Battery disposal contaminating soil in some African cases
 - Biomass systems potentially depleting local resources
- **Best Practices**
 - Site-specific environmental assessments
 - Community engagement in planning
 - Circular economy approach includes battery recycling programs, Component refurbishment and Sustainable material sourcing [1]

Chapter 6

Conclusion and recommendation for Operational Sustainability of Mini grid

The proposed 5.2 kW solar mini-grid for Rajapuri village is technically feasible and economically viable (LCOE: ₹5.82 per kWh). It balances cost-effectiveness with reliability, though HOMER Pro's higher-capacity design (10.4 kW) offers greater resilience against weather variability.

Sustainability Recommendations for Rajapuri Mini-Grid

- **Adaptive System Design**
 - Size capacity 20–30% above current demand
 - Modular architecture for easy expansion
- **Community Empowerment**
 - Train local “Energy Champions” for daily operations and basic repairs
 - Establish village oversight committees
- **E-Waste Solutions**
 - Formal partnerships with certified recyclers
 - Battery buy-back programs
 - Periodic collection drives
- **Smart Monitoring**
 - IoT sensors for performance tracking and predictive maintenance
 - Cloud-based analytics dashboard

Replicability: By integrating technology with grassroots governance, Rajapuris mini-grid model can become a replicable model for rural electrification in India.

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<u>Table 6.1: Team Contribution</u>	
Roll Number	Contribution
24m1345	25 %
24m1346	25 %
24m1349	25 %
24m1350	25 %